Rishee Deshmukh

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Professional Summary

Full-stack software engineer with a Master's in Computer Science from Syracuse University and hands-on experience across backend services, full-stack platforms, AI-driven systems, cryptographic protocol design, and distributed systems. Brings strong foundations in Python, Go, JavaScript/TypeScript, cloud infrastructure, and API design, with a track record of shipping technically rigorous projects from concept to deployment.

Education

Syracuse University – *Master of Science, Computer Science (GPA: 3.47/4.0)* *May 2026*
Medi-Caps University – *Bachelor of Technology, Computer Science and Engineering* *May 2024*

Experience

Backend Engineer (AWS) Jul 2026 – Present
ZenZiee Corp. *Remote*

- Refactored DynamoDB table schemas to eliminate redundant data storage and implemented query pagination using Global Secondary Indexes, reducing read throttles and improving data retrieval efficiency
- Conducted AWS infrastructure cost analysis across ECS Fargate, ElastiCache, DynamoDB, and networking services, identifying approximately 50% cost reduction through right-sizing and provisioning optimization
- Replaced a hand-rolled salt-and-concatenate bcrypt scheme with standard bcrypt hashing (cost 12) using a unique random per-user salt, with lazy rehash-on-login so existing accounts upgrade automatically, built, tested and submitted for review in 4 days at the CEO's request
- Developed backend services in Go with DynamoDB, ElastiCache Redis, ECS Fargate, and WebSocket integration on AWS

Full-Stack Developer Nov 2025 – May 2026
iConsult Collaborative, Syracuse University *Syracuse, NY*

- Designed and developed full-stack web application modules using React and Next.js in a TypeScript codebase
- Evaluated system design trade-offs across frontend frameworks to improve modularity and maintainability
- Implemented CI/CD pipelines via GitHub Actions within Agile development cycles for reliable delivery

Network Engineering Intern May 2023 – Jul 2023
Cisco *Indore, India*

- Configured and deployed switches, routers, and end devices for reliable connectivity across local and remote environments
- Designed IPv4/IPv6 addressing schemes supporting scalable network segmentation

Projects

Multi-Agent Code Review System (LangGraph / FastAPI) Jan 2026 – May 2026

- Architected a parallel multi-agent review pipeline using LangGraph fan-out/fan-in topology with three specialized agents (Security, Performance, Maintainability) and a Claude Sonnet synthesizer for deduplication and gap analysis
- Achieved 90% recall (27/30 vulnerabilities) across 9 CWE classes combining Semgrep, Bandit, Ruff, and ESLint static analysis with LLM-based semantic reasoning
- Built full-stack platform with FastAPI, React, and Supabase; implemented SSE streaming for progressive result delivery at 52.3s average processing time

SecureChat: End-to-End Encrypted Messaging Protocol (Python) Aug 2025 – Dec 2025

- Built a client-server encrypted chat application implementing X25519 Diffie-Hellman key exchange and ChaCha20-Poly1305 AEAD with HKDF-SHA256 key derivation
- Designed replay attack prevention using Bloom filter-based message ID tracking and timestamp validation with TTL
- Implemented Signal-inspired key ratcheting deriving fresh encryption keys per message for forward secrecy

E-Ballot: Blockchain-Based Voting System Sept 2023 – Dec 2023

- Architected a decentralized e-voting system on Ethereum using three Solidity smart contracts optimized for gas efficiency; Voting Contract consumed 734,353 gas per deployment
- Achieved 66.67 votes/minute throughput, outperforming traditional voting systems by 4.3x as validated in a peer-reviewed publication
- Integrated Web3.js with MetaMask for trustless wallet-based voter authentication with real-time on-chain vote verification

Transformer-Based Automatic Text Simplification (T5 / HuggingFace) Aug 2025 – Dec 2025

- Fine-tuned T5-small on 50,000 WikiAuto sentence pairs; benchmarked against BART and Phi-3 Mini (zero-shot) on ASSET
- Built an end-to-end ML pipeline with mixed-precision (FP16) training, gradient accumulation, and multi-metric evaluation (SARI, BLEU, ROUGE-L, Flesch Reading Ease)
- Analyzed trade-offs between reference-based metrics and readability, finding zero-shot LLMs achieved highest Flesch score (59.6) while supervised models maximized BLEU/ROUGE through conservative edits

Password Manager (C++ / Qt / CMake / libsodium) Aug 2026

- Built a C++ desktop password manager with a Qt-based GUI, implementing full credential CRUD operations and encrypted local storage
- Implemented master-password protection and secure password generation using libsodium cryptographic primitives
- Managed the build system with CMake, applying hands-on memory management and object-oriented design in a security-sensitive application

SKILLS

Languages: Python, Go, Java, JavaScript, TypeScript, C, C++, C#, Solidity
Backend: Node.js, Express.js, FastAPI, Django, Kafka, REST APIs, API Design, WebSockets, Server-Sent Events
Frontend: React, Next.js, HTML5, CSS3, Tailwind CSS
Databases: PostgreSQL, DynamoDB, MySQL, MongoDB, Redis, Supabase, Oracle
Cloud/DevOps: AWS (ECS Fargate, ElastiCache), GCP, Docker, Kubernetes, Git, CI/CD, GitHub Actions, CMake
AI/ML: TensorFlow, PyTorch, Scikit-learn, HuggingFace, LangGraph, LLM Integration, NumPy
Security/Networking: Applied Cryptography (X25519, ChaCha20-Poly1305, HKDF), bcrypt, libsodium, Semgrep, Bandit, Cisco IOS, IPv4, IPv6, VLAN, ACLs
Blockchain: Ethereum, Smart Contracts, Web3.js
Desktop: Qt